

SegmentGDA: Training-Free Few-Shot Segmentation via Gaussian Discriminant Analysis on Vision Foundation Features

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Motivation

Segment a novel object from only a few annotated support examples.

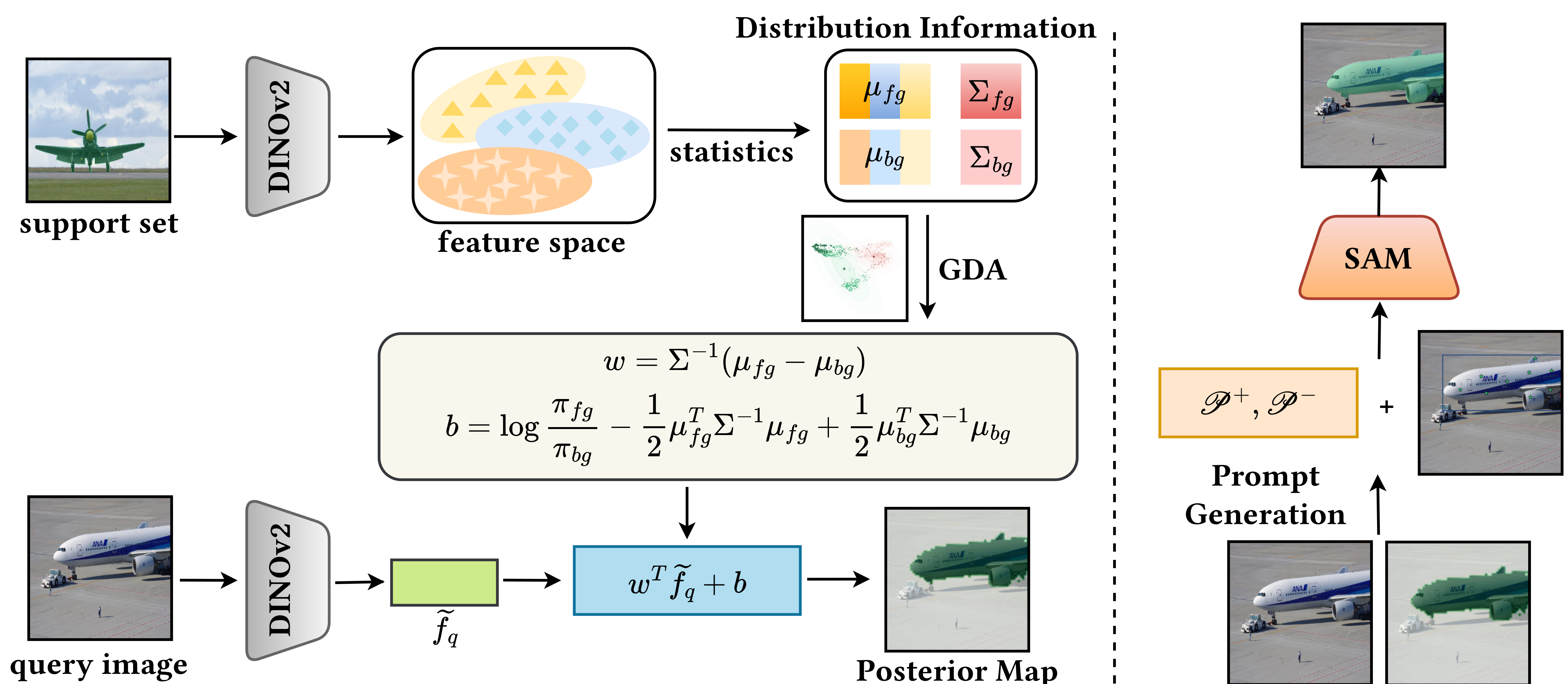
Local patch matching can miss object parts and select background clutter.

Method overview

Core idea

Model foreground and background as distributions in frozen foundation features.

Bayesian posteriors guide SAM prompts. No task-specific training or fine-tuning.



1 Frozen features

DINOv2 ViT-L/14 features, signed square-root transform and L2 normalization.

2 Gaussian inference

Two class means and a shared covariance yield patch-level foreground posteriors.

3 SAM refinement

Diverse positive and negative points guide frozen SAM ViT-H to iteratively correct errors.

5-shot comparison with training-free methods

mIoU (%) Selected baselines from Tables 1-2 of the paper

Method	COCO-20 ⁱ	FSS-1000	LVIS-92 ⁱ	PASCAL-5 ⁱ
Matcher	60.7	89.6	40.0	74.0
UNICL-SAM	78.7	86.3	37.4	-
GF-SAM	66.8	88.9	44.2	82.6
SegmentGDA (ours)	62.5	89.8	41.8	82.2

Bold: best among the methods shown. -: not reported in our paper.

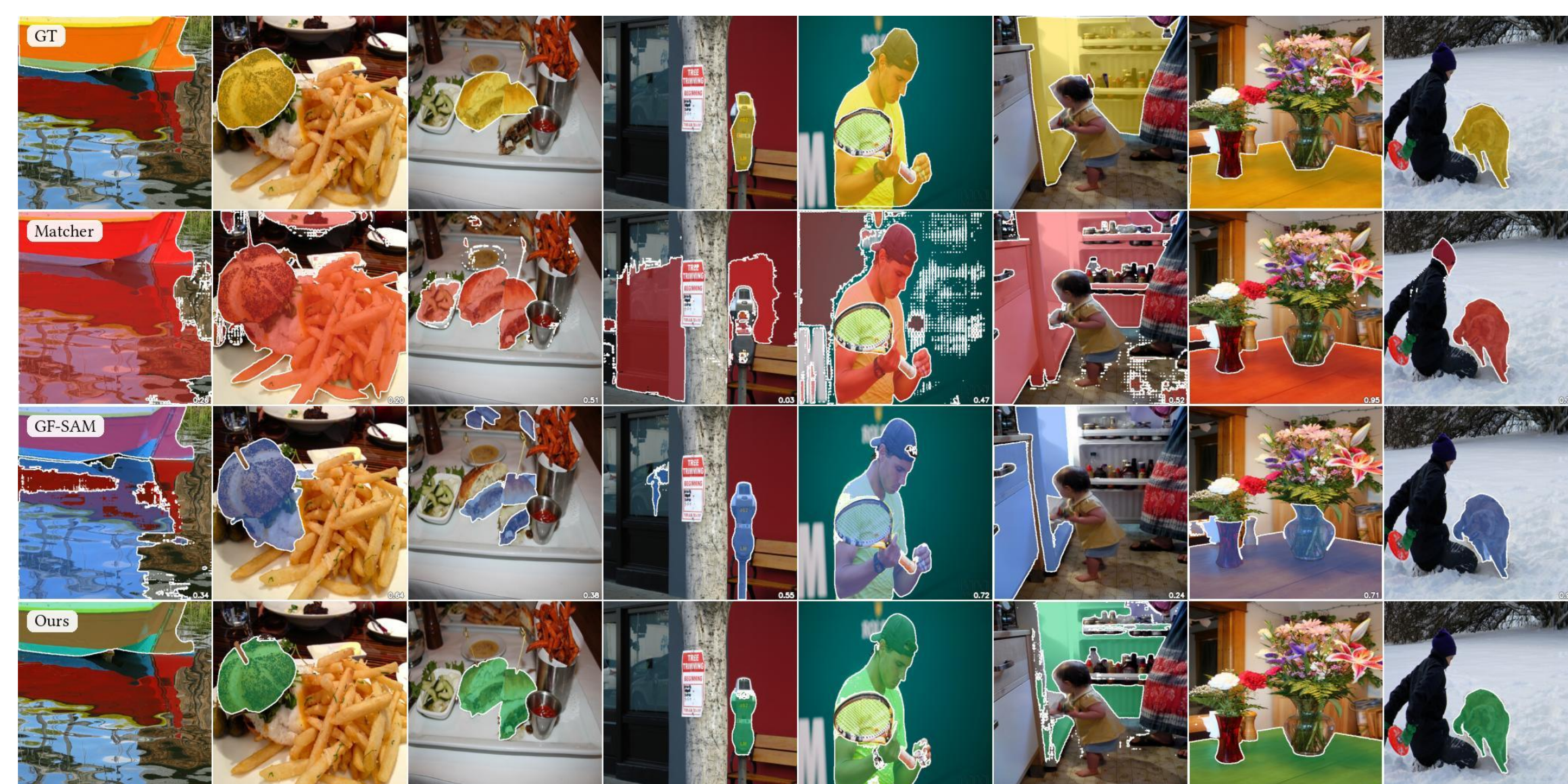
Our 1-shot mIoU, in the same dataset order: 42.1 / 82.9 / 26.2 / 59.5.

Higher than Matcher on all four benchmarks; competitive on FSS-1000 and PASCAL-5ⁱ.

Single-example performance remains a limitation; the main gains occur with multiple supports.

Qualitative comparisons

COCO-20ⁱ



Rows: ground truth / Matcher / GF-SAM / SegmentGDA. Selected 5-shot episodes.

FSS-1000



TAKEAWAY Global statistics improve multi-shot mask transfer.

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CODE